

Quantum Computing

Noah Jaye

Cows

- 5 children, ranked by favoritism
- 20 cows
- One shot
- How to split?



Pebbles

- You have at least 20 pebbles
- What if pebbles $\mapsto$ cows?
- Computation



Classical computing

- Modern classical computing is computing that use **binary digits**, or **bits**, represented by nanoscale “switches” that are either on or off, 0 or 1.
- Multiple bits can be used to store more complicated information, since two bits can be arranged in four ways (00, 01, 10, 11), as opposed to two for a single bit (0, 1).
- Classical computers manipulate bits according to given instructions, which is the basis of all modern computing.

Classical limitations

- Classical computers are very good at quickly completing many problems. However, certain problems take significant time to compute, even for classical computers.
- Large unstructured search problems are very difficult, even for computers.



Cows

- Cow lineup competition
- Each cow is assigned a unique power of 2 (1, 2, 4, 8, etc.)
- Each cow's number is multiplied by its position in the line.
- Numbers all added up
- If the sum is equal to 17 170 415, they win a prize



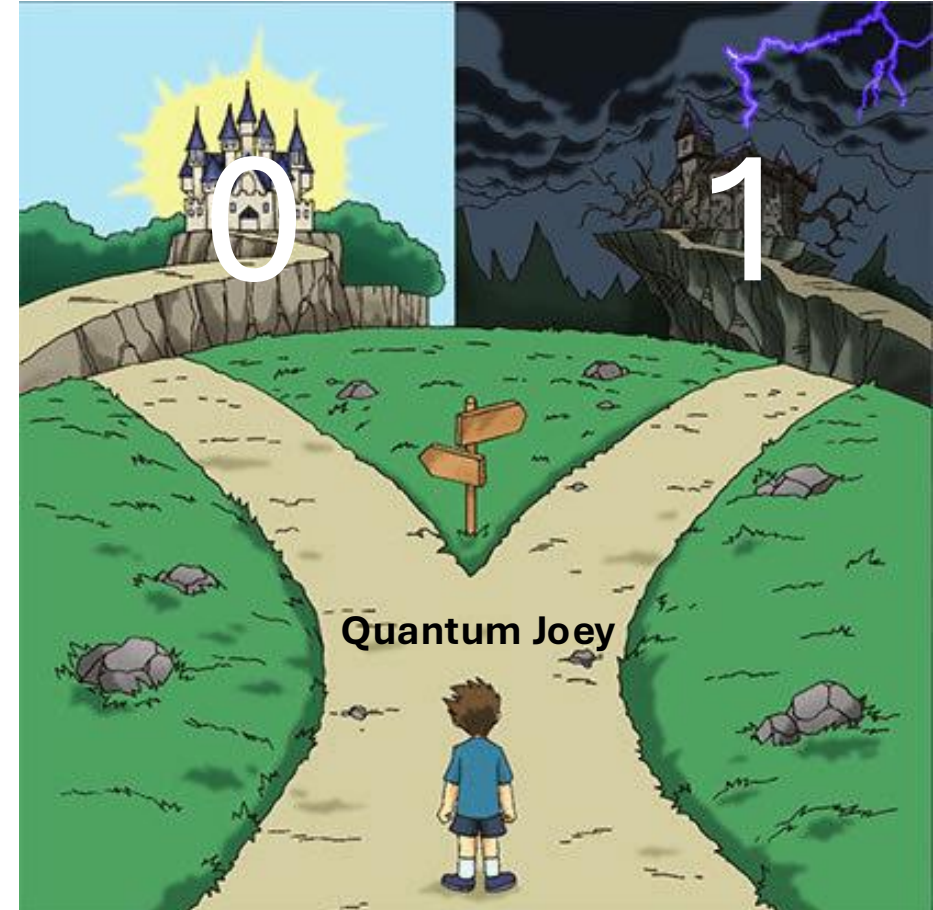
Quantum effects

- Superposition
- Entanglement



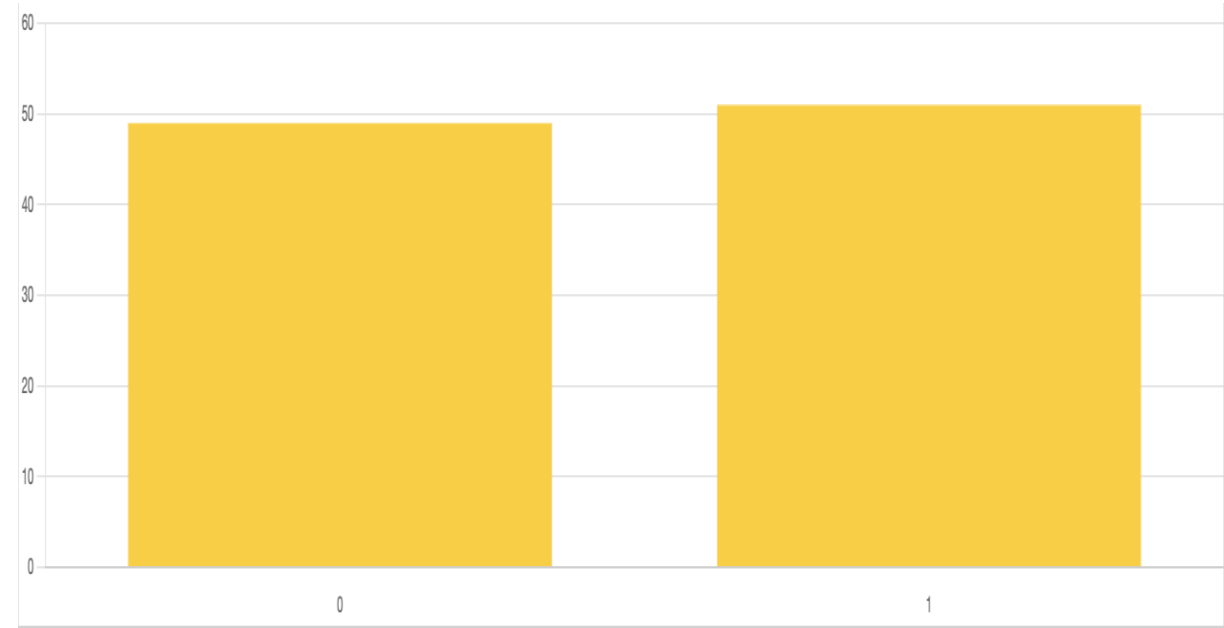
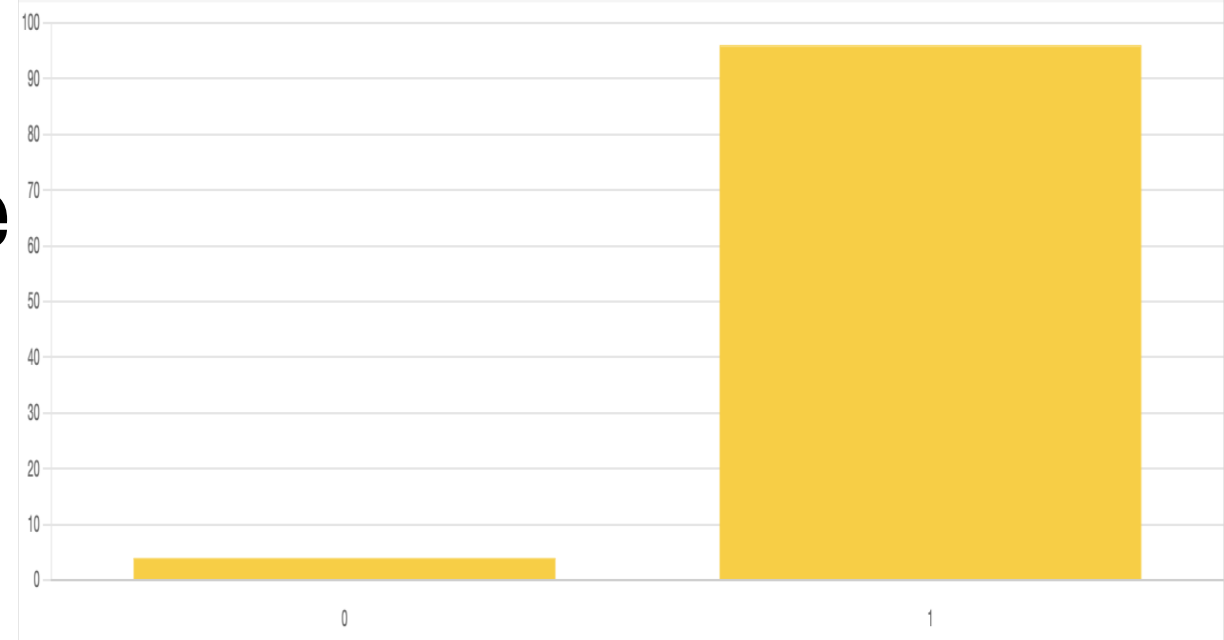
Superposition

- Any quantum particle exists in multiple states at once, and ‘picks’ one randomly when interacted with.
- Particles can be more likely to exist in certain states, though.
- This means that when you interact with the particle, there is a higher chance that it appears as one state than another.



Superposition example

- Try to match the following particles to their distributions
- Particle A: 50/50
- Particle B: 90/10

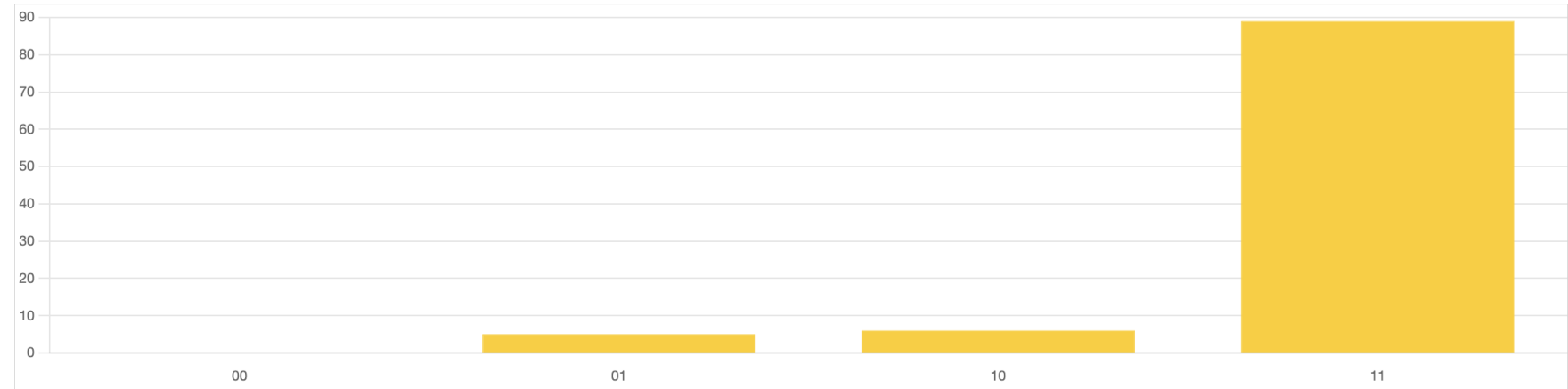
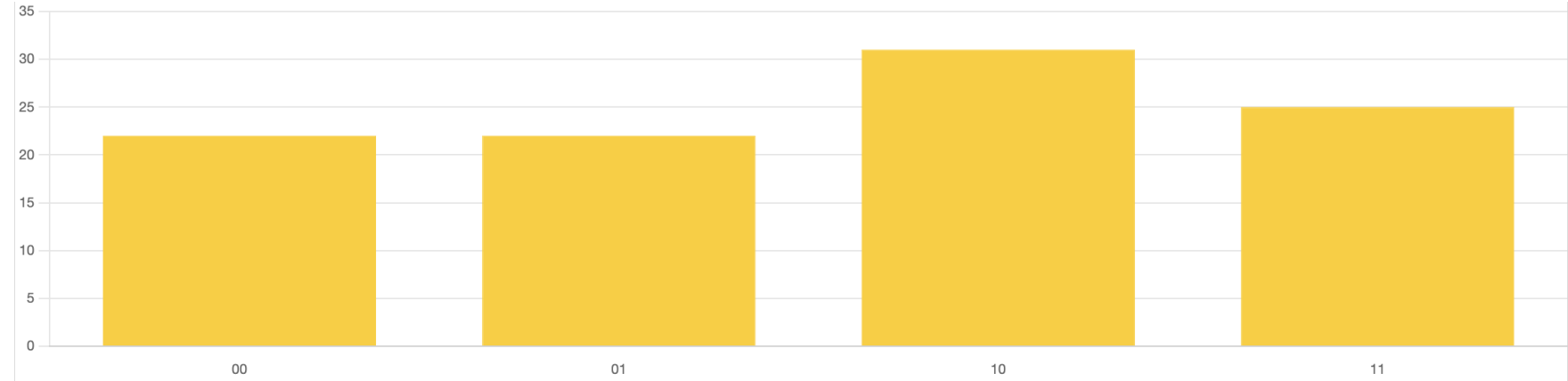


Entanglement

- Due to magic, quantum particles in superposition can become ‘entangled’.
- This means that when you interact with particle A, if it is entangled with particle B, particle B will pick a state as if it was interacted with as well.
- When particles are entangled, the group of particles has probabilities to be in certain states, just like superposition. Until observed, the group exists in every possible combination of states.

Entanglement example

- Here are two entangled states



The qubit

- A qubit is any quantum particle used for computation that has two states.
- Because quantum particles can exist in exponentially many states at once, a handful of qubits can perform many computations at once.
- Quantum computers work by harnessing these entanglements.



Qubits as vectors

- Qubits can be represented by column vectors:

- $0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, 1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

- $0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, 1 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, 2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, 3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

- These are “basis vectors”

Qubits as vectors

- Qubits can be represented by column vectors:

- $|+\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, |-\rangle = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$

- These qubits are in superposition. When observed, their entries are squared. The value at each entry is the probability that the qubit will be observed in that state.
- The sum of these probabilities must be 1

Qubits as vectors

- These qubits are in superposition. When observed, their entries are squared. The value at each entry is the probability that the qubit will be observed in that state.

- $$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}^2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \quad \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} \end{bmatrix}^2 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

- Both of these qubits are in a 50/50 superposition.

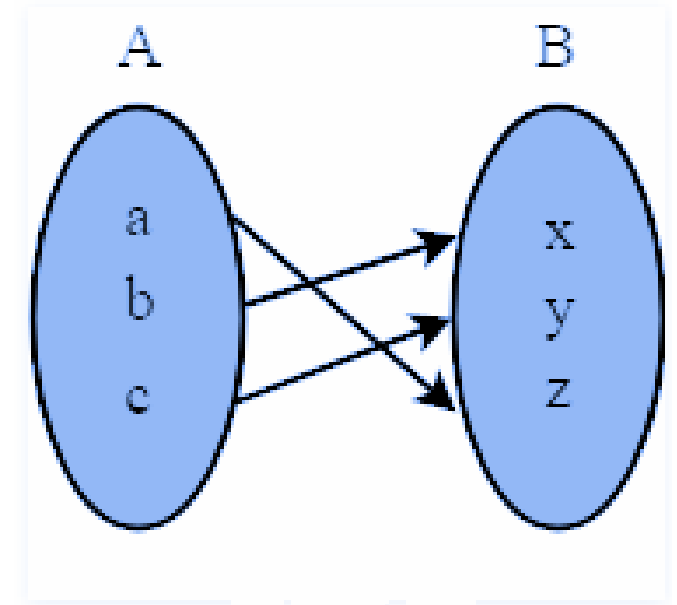
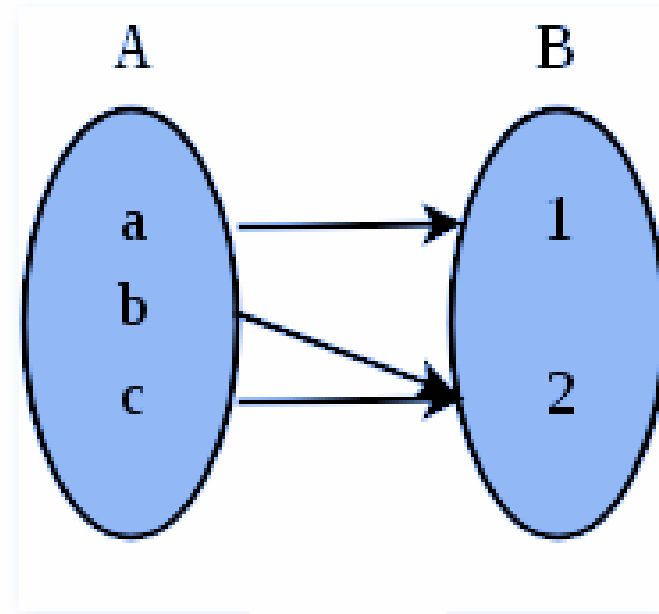
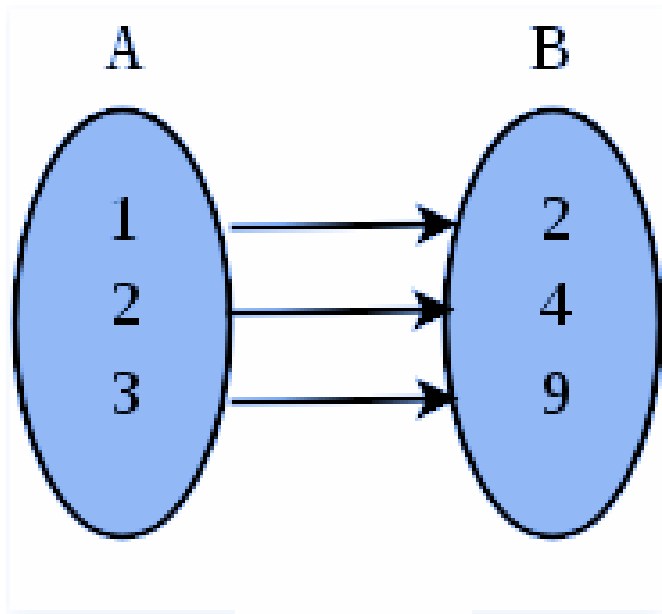
Quantum operators

- Quantum operators are how we manipulate qubits. They have to do a few things, which basically boil down to preserving information.

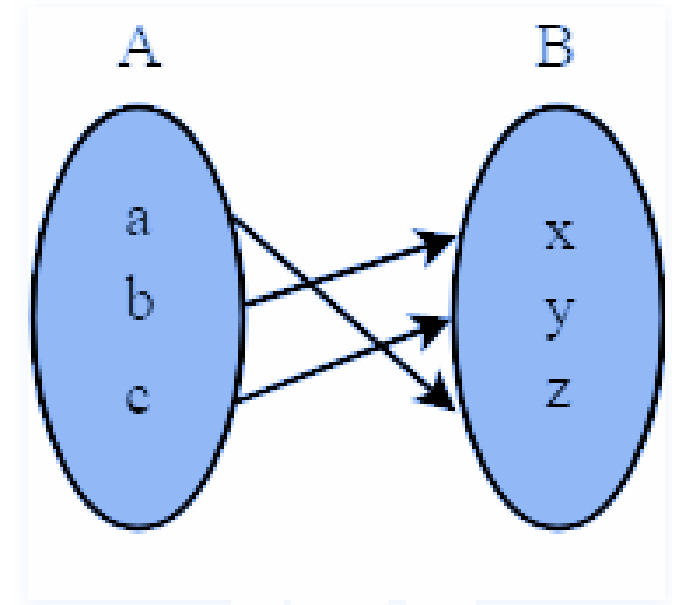
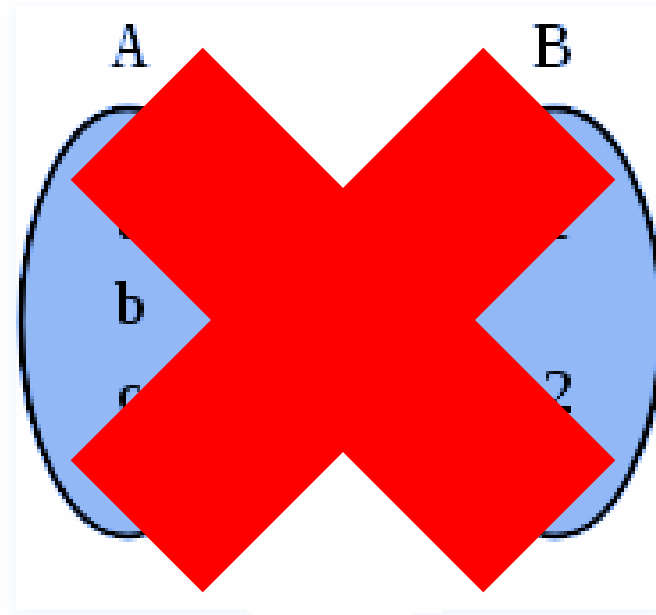
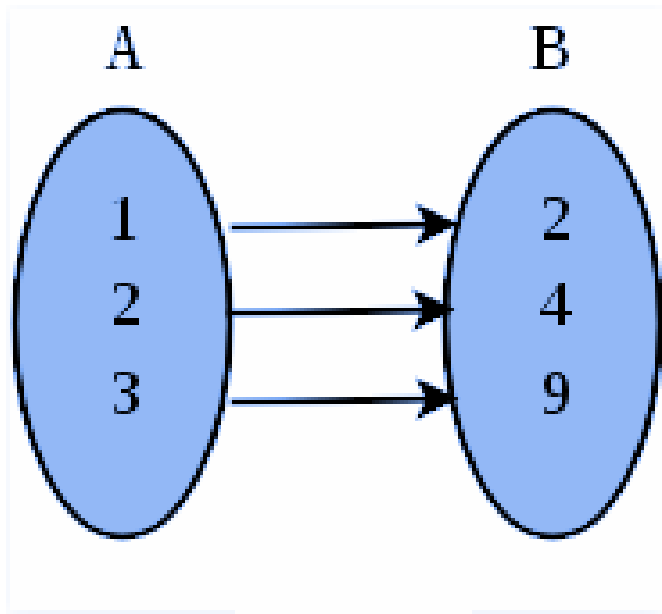
Quantum operators

- Quantum operators are how we manipulate qubits. They have to do a few things, which basically boil down to preserving information.
- 1: When we apply a quantum operator to a qubit, we need to be able to reverse the operation.
- This means that every input must have a unique output.

Quantum operators



Quantum operators

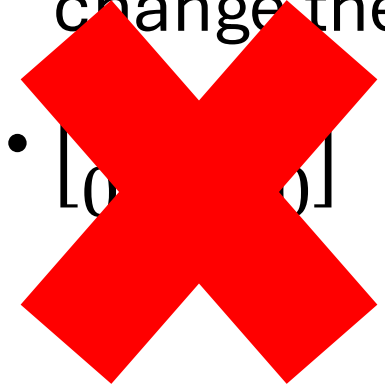


Quantum operators

- Quantum operators need to maintain a total probability of 1.
- An operator that doubles a qubit doesn't work, because it would change the total probability.
- $\begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

Quantum operators

- Quantum operators need to maintain a total probability of 1.
- An operator that doubles a qubit doesn't work, because it would change the total probability.



• $|0\rangle \rightarrow |0\rangle$

Quantum operators

- Quantum operators can be represented by square matrices that are **unitary**.
- Unitary is a fancy word to describe the restrictions on quantum operators.
- To check if an operator is unitary:

Quantum operators

- Quantum operators can be represented by square matrices that are **unitary**.
- Unitary is a fancy word to describe the restrictions on quantum operators.
- To check if an operator is unitary:
 1. There needs to be a matrix that reverses what it does, or an “inverse” of the matrix.

Quantum operators

1. There needs to be a matrix that reverses what it does, or an “inverse” of the matrix.

First, let's try $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$.

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 * 1 + 1 * 0 \\ 0 * 1 + 0 * 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

However,

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 * 0 + 1 * 1 \\ 0 * 0 + 0 * 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Two inputs for the same output!

Quantum operators

1. There needs to be a matrix that reverses what it does, or an “inverse” of the matrix.

Next, let's try $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 * 1 + 1 * 0 \\ 1 * 1 + 0 * 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

However,

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 * 0 + 1 * 1 \\ 1 * 0 + 0 * 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Different outputs!

Quantum operators

- Quantum operators can be represented by square matrices that are **unitary**.
- Unitary is a fancy word to describe the restrictions on quantum operators.
- To check if an operator is unitary:
 - 1. There needs to be a matrix that reverses what it does, or an “inverse” of the matrix.
 - 2. The matrix needs to be symmetrical along its top left to bottom right diagonal.

Cows

- Now that you have a quantum computer, how to win?



Grover's algorithm

- Instead of having to check n items, it checks $\sqrt{n}$ times, which is much faster
- Good example of what quantum computers can and can't do.



Grover's algorithm

- First, set up an equal superposition of all of your bits.
- For three qubits, this means 8 states with value $1/\sqrt{8}$.



$$\left[\begin{array}{c} 1 \\ \sqrt{8} \\ 1 \\ \sqrt{8} \\ 1 \\ \sqrt{8} \\ 1 \\ \sqrt{8} \\ 1 \\ \sqrt{8} \\ 1 \\ \sqrt{8} \end{array} \right]$$

Grover's algorithm

- Then, flip the sign of your correct bit.
- Because operators must be unitary, we can't just find the right answer here.

[illegible]

Grover's algorithm

- After that, find the average of all values. This is equal to $\frac{(n-2)(\frac{1}{\sqrt{n}})}{n}$, or $\frac{6}{8\sqrt{8}}$
- Double this value, and for each entry, subtract that entry from the doubled average.

$$\bullet \frac{12}{8\sqrt{8}} - \frac{1}{\sqrt{8}} = \frac{4}{8\sqrt{8}}, \text{ or } \frac{1}{2\sqrt{8}}$$

$$\bullet \frac{12}{8\sqrt{8}} - \frac{-1}{\sqrt{8}} = \frac{20}{8\sqrt{8}}, \text{ or } \frac{5}{2\sqrt{8}}$$

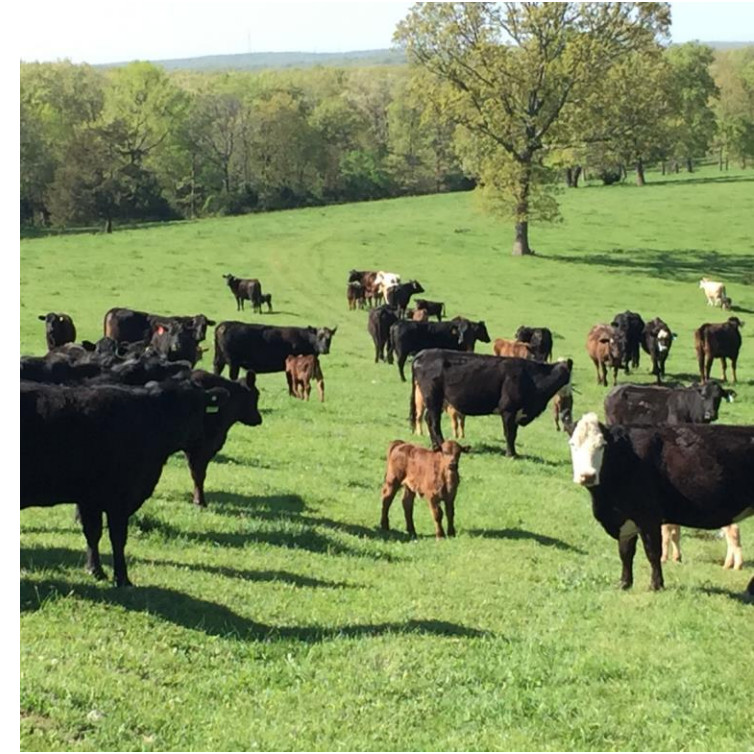
[illegible]

Grover's algorithm

- When we observe this state, there is a high chance that we find the correct answer.
- $\left(\frac{5}{2\sqrt{8}}\right)^2 = \frac{25}{32}$, or a 78% chance of a correct answer.
- Percent approaches 100 as problem gets bigger and $\sqrt{n}$ approaches ∞ .

Grover's algorithm ft. cows

- There are $20!$ arrangements of cows, or $20 * 19 * 18 \dots * 2 * 1$.
- This is $2.4329E+18$.
- $\sqrt{n}$ is ~ 1.5 billion in this case, and after that many iterations, correct answer is $\sim 100\%$.

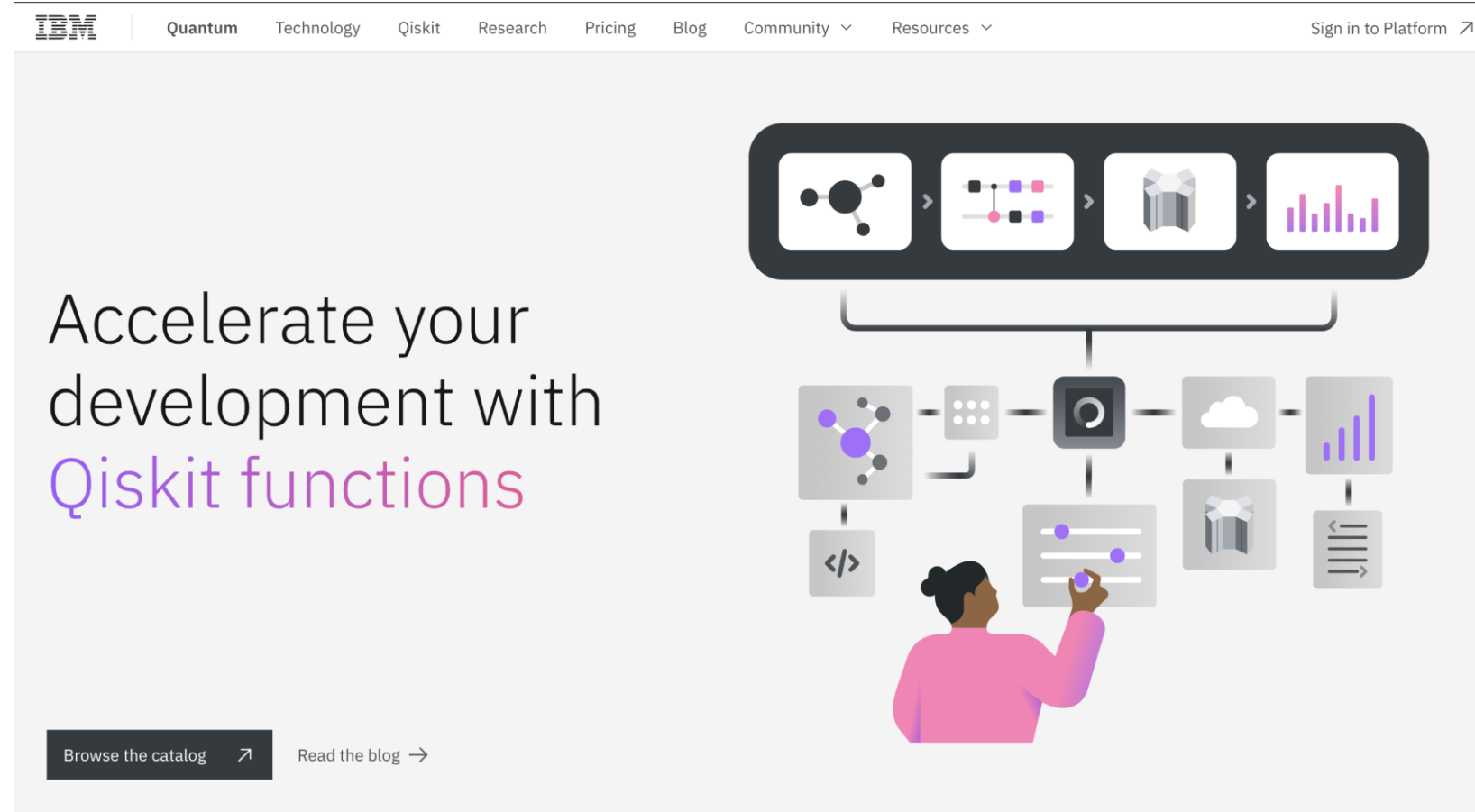


Classical computing redux: Wait why don't we use real precision bits

- Imagine you have a copper wire. $\frac{V}{I} = R$, and so we should be able to perform infinitely precise multiplication by changing current and measuring voltage, and vice versa.
- In real life, precision and error are real concerns that quantum computers aren't immune to.
- Quantum computers struggle with error and precision.

IBM

- <https://quantum.ibm.com/composer/files/new>
- IBM



Sources

- <http://thequantumlaend.de/quantum-circuit-designer/>
- <https://byjus.com/maths/bijective-function/#:~:text=What%20is%20Bijective%20Function%3F,a%E2%80%9D%20in%20the%20domain%20A.>
- <https://www.ibm.com/quantum>

Image sources

- [Wallpaper : animals, grass, field, farm, grassland, pasture, herd, agriculture, meadow, plain, fauna, prairie, rural area, habitat, natural environment, cattle like mammal, grazing, dairy cow 2560x1600 - Kotan - 248243 - HD Wallpapers – WallHere](#)
- [https://images.pexels.com/photos/173757/pexels-photo-173757.jpeg?cs=srgb&dl=pexels-ciboulette-173757.jpg&fm=jpg](#)
- [https://images.freeimages.com/images/large-previews/f31/free-cow-image-moo-cows-in-a-line-in-a-field-1631828.jpg](#)
- [https://th.bing.com/th/id/OIP.3MFqdKdOcuFRQ6TwAmtP3wHaKr?rs=1&pid=ImgDetMain](#)
- [https://deepstash.com/_next/image?url=https:%2F%2Fd1dfxfqogsjxt.cloudfront.net%2F150388-1633255836.jpg&w=828&q=75](#)